Lecture 8

- KF independence assumptions
- Markovianity, cond. indep. measurements
- Linear structure of jointly Gauss. R.V.s

Recall to derive the KF eqns we required assumptions:

A1: V_t indep. of $Y_{1:t-1}$ $\forall t$
A2: V_t indep of X_t $\forall t$
A3: W_t indep of $Y_{1:t-1}$ $\forall t$
A4: W_t indep of X_t $\forall t$

Recall L-G SS model:

$$X_{t+1} = A_t X_t + B_t u_t + W_t,$$
(a1:) $W_t \sim GWN(0, Q_t)$
$$Y_t = C_t X_t + V_t,$$
(a2:) $V_t \sim GWN(0, R_t)$
(a3:) V_t, W_τ indep $\forall t, \tau$
(a4:) V_t, W_t indep of X_0 $\forall t$

Claim:
$$a1 - a4 \implies A1 - A4$$
We'll prove each of A4, ..., A1 below

A4: W_t indep of X_t $\forall t$
from L-G SS model:

$$\begin{aligned} X_t &= A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1} \\ &= A_{t-1}A_{t-2}X_{t-2} + B_{t-1}u_{t-1} + A_{t-1}B_{t-2}u_{t-2} \\ &\quad + W_{t-1} + A_{t-1}W_{t-2} \\ &= \Big(\prod_{\tau=0}^{t-1} A_\tau\Big)X_0 + \sum_{\tau=0}^{t-1}\Big(\underbrace{\prod_{s=\tau+1}^{t-1} A_s}_{\text{for } s=t \Rightarrow I}\Big)B_\tau u_\tau \\ &\quad + \sum_{\tau=0}^{t-1}\Big(\underbrace{\prod_{s=\tau+1}^{t-1} A_s}_{\text{for } s=t \Rightarrow I}\Big)W_\tau \end{aligned}$$

$$X_\tau = \Phi_{t-1}X_0 + \begin{bmatrix}\Psi_{t-1} & \cdots & \Psi_0\end{bmatrix}\begin{bmatrix} B_{t-1}u_{t-1} \\ \vdots \\ B_0 u_0\end{bmatrix} + \begin{bmatrix}\Psi_{t-1} & \cdots & \Psi_0\end{bmatrix}\begin{bmatrix} W_{t-1} \\ \vdots \\ W_0\end{bmatrix}$$

so if W_t is indep. of $W_{0:t-1}$
(a2) $\implies$ so $W_{0:t}$ is white (i.e. W_t indep W_τ $\forall t \neq \tau$)
and if W_t is indep. of X_0 (a4)
then A4 holds.

$\boxed{(a2, a4) \implies A4}$ □

A3: W_t indep. of $Y_{1:t-1}$

$$Y_\tau = C_\tau X_\tau + V_\tau$$
$$X_\tau = \Phi_{\tau-1}X_0 + \Psi_{0:\tau-1}W_{0:\tau-1} + \Psi_{0:\tau-1}(Bu)_{0:\tau-1}$$

so Y_τ depends on $X_0, W_{0:\tau-1}$ and V_τ
so if W_t indep. of X_0 (a4)
and W_t indep. of $W_{0:\tau-1}$ $\tau < t$ (a2)
and W_t indep. of V_τ, $\tau < t$ (a3)

so

$\boxed{(a2, a3, a4) \implies A3}$ □

A2: V_t indep. of X_t $\forall t$

$$X_t = \Phi_{t-1} X_0 + \Psi_{0:t-1} (Bu)_{0:t-1} + \Psi_{0:t-1} W_{0:t-1}$$

so if V_t is indep of X_0 (a4)
and V_t is indep of $W_{0:t-1}$ (a3)

$$\boxed{(a3, a4) \Rightarrow A2}$$ □

A1: V_t indep. $Y_{1:t-1}$ $\forall t$

$$Y_\tau = C_\tau X_\tau + V_\tau \quad \forall \tau < t$$
$$X_\tau = \Phi_{\tau-1} X_0 + \Psi_{0:\tau-1} W_{0:\tau-1} + \Psi_{0:\tau-1} (Bu)_{0:\tau-1}$$

so if V_t indep. of X_0 (a4)
and V_t indep. of $W_{0:\tau-1}$ (a3)
and V_t indep of V_τ (a1) $\forall \tau < t$
so $\boxed{(a1, a3, a4) \Rightarrow A1}$ □

Linear Structure of Jointly Gaussian R.V.s

Consider $(X, Y) \sim N(\mu, \Sigma)$
let $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$
Turns out

Thm: $\exists C \in \mathbb{R}^{m \times n}$ and R.V. V, $v \in \mathbb{R}^m$ s.t.

$$\boxed{Y = CX + V}$$

where $V \sim N(\mu_v, R_v)$
and $\boxed{V \text{ indep of } X}$

intuition: C projects X into space of Y, gives all info about Y that can be obtained from X.
V is the leftovers, which are independent of X.

also $\exists M \in \mathbb{R}^{n \times m}$ and R.V. W, $w \in \mathbb{R}^n$ s.t.

$$\boxed{X = MY + W}$$

where $W \sim N(\mu_w, Q_w)$
and $\boxed{W \text{ indep of } Y}$

Recall Gaussian Est. Eqns:

$$\mu_{x|y} = \mu_x + \Sigma_{xy} \Sigma_y^{-1} (y - \mu_y)$$
$$\Sigma_{x|y} = \Sigma_x - \Sigma_{xy} \Sigma_y^{-1} \Sigma_{yx}$$

claim:

$$\boxed{M = (\Sigma_{xy} \Sigma_y^{-1})}$$

and $\boxed{\mu_w = (\mu_x - \Sigma_{xy} \Sigma_y^{-1} \mu_y)}$

then we get

$$E[X|Y=y] = My + E[V]$$
$$My + \mu_w = \mu_{x|y}$$

and $\boxed{Q_w = \Sigma_{x|y}}$

Now consider state and measurement sequence jointly Gaussian:

$$(X_{0:t+1}, Y_{1:t}) \sim N(\mu, \Sigma)$$

By linearity of jointly Gaussian R.V.s we know:

$$X_{t+1} = A_t X_t + \Phi_{0:t-1} X_{0:t-1} + \Psi_{1:t} Y_{1:t} + \overline{W}_t$$

where $\overline{W}_t \sim N(\mu_{w_t}, Q_{w_t})$
and $\overline{W}_t$ indep of $X_{0:t}$, $Y_{1:t-1}$

to get Markovianity we require X_{t+1} given X_t indep $X_{0:t-1}$, $Y_{1:t}$
$\Rightarrow$ require $\Phi_{0:t-1}, \Psi_{1:t} = 0$

to get

$$X_{t+1} = A_t X_t + \overline{W}_t$$

let's redefine

$$\overline{W}_t = B_t u_t + W_t, \quad W_t \sim N(0, Q_t)$$
(so $\mu_{w_t} = B_t u_t$)

$$\boxed{X_{t+1} = A_t X_t + B_t u_t + W_t} \quad W_t \sim N(0, Q_t)$$

so jointly Gaussian
+ Markov $\Rightarrow$ SS dynamics

Similarly: jointly Gaussian
+ Cond. indep. meas.
$\Rightarrow$ SS meas. eqn.

$(X_{0:t}, Y_{1:t}) \sim N(\mu, \Sigma)$
by linearity of jointly Gaussian R.V.

$$Y_t = C_t X_t + \Theta_{0:t-1} X_{0:t-1} + K_{1:t-1} Y_{1:t-1} + \bar{V}_t$$

where $\bar{V}_t \sim N(\mu_{V_t}, R_{V_t})$
$\bar{V}_t$ indep $X_{0:t}, Y_{1:t-1}$

so for cond. indep. meas.
I require Y_t given X_t is indep. of $X_{0:t-1}, Y_{1:t-1}$

$\Rightarrow \Theta_{0:t-1} = 0$
and $K_{1:t-1} = 0$

we get

$$Y_t = C_t X_t + \bar{V}_t$$

redefine

$$\bar{V}_t = D_t u_t + V_t$$
$$V_t \sim N(0, R_t)$$

to get

$$\boxed{Y_t = C_t X_t + D_t u_t + V_t}$$

SS model!

Take home part:

The L-G SS model is the only model that is:

(i) jointly Gaussian w/ $X_{0:t}, Y_{1:t}$
(ii) Markov
(iii) Cond. Ind. Measurements